

Three-Day Workshop

Theme: ICT-Based Research, Innovation, and Scientific Paper Writing for Professional Engineers

Target Participants: **Professional Engineers**

Total Participants: 30

Duration: 3 Days (15-17/02/2083)

Time: 10:00 AM – 5:00 PM

Tutor/Guide: **Pradip Paudyal, PhD (Engineering)**

Expected Learning Outcomes

By the end of the workshop, participants will be able to:

- ✓ Identify ICT research problems and innovation opportunities.
- ✓ Conduct effective literature reviews.
- ✓ Design appropriate research methodologies.
- ✓ Prepare academic and grant-oriented research proposals.
- ✓ Present and defend research ideas professionally.
- ✓ Apply innovation-driven approaches in ICT research projects.

Workshop Design

To make the workshop highly effective, the workshop will follow collaborative group-based learning model and guided mentoring and peer review approach.

Participants will work in 5/6 groups of 5 participants each throughout the workshop.

<u>Groups Task</u>	<u>Group Structure</u>
✓ Select one ICT-based research problem ✓ Conduct literature review collaboratively ✓ Design a methodology ✓ Develop a mini research proposal ✓ Draft sections of a scientific paper ✓ Present final outputs on Day 3	Each group of 5 participants may assign roles: ✓ <i>Group Coordinator</i> — manages workflow and presentation ✓ <i>Literature Review Lead</i> — searches and summarizes papers ✓ <i>Methodology Lead</i> — develops technical approach ✓ <i>Writing Lead</i> — drafts proposal/paper sections <i>Presentation Lead</i> — prepares final presentation slides

Tools: Google Scholar, SJR, Mendeley, Overleaf, MATLAB/Python/R

Day 1: Research Problem Identification and Literature Review

Theme: *From Engineering Challenges to Research Questions*

	Session	Type /Method	Topics / Activities
10:00 – 10:20 AM	Opening & Workshop Orientation	Introductory /Interactive	Workshop goals, expected outputs, group formation, introduction to ICT research ecosystem
10:20 – 11:05 AM	Session 1: <i><u>ICT Research and Innovation Landscape</u></i>	Theory/Lecture + Discussion	Research, innovation, scientific research, ICT based innovation, innovation ecosystem, research area and topic selection
11:05 AM – 12:15 PM	Session 2: Group Activity — Research Problem Identification	Hands-on/Group Work	Each group identifies problems and converts them into researchable topics
12:15 – 1:15 PM	Lunch Break		
1:15 – 2:30 PM	Session 3: <i><u>Literature Review and Research Gap Analysis</u></i>	Theory/Demonstration + Discussion	Literature review process, using SJR, Google Scholar and IEEE Xplore, identifying research gaps, reference management, resources/databases
2:30 – 2:45 PM	Tea/Coffee Break		
2:45 – 4:15 PM	Session 4: Group Literature Review Workshop	Hands-on/Guided Group Practice	Groups search papers, extract gaps, summarize findings, prepare references
4:15 – 5:00 PM	Group Mentoring & Research Topic Validation	Interactive/Trainer Rotation	Trainer visits each group for feedback, topic refinement, and feasibility discussion

Deliverables:

Research title/topic, problem statement, initial literature review summary and identified research gap

Day 2: Methodology Design and Scientific Paper Writing

Theme: *Designing Research and Scientific Paper Writing*

	Session	Type /Method	Topics / Activities
10:00 – 10:20 AM	Day 1 Review & Group Presentations	Interactive /Group Sharing	Each group briefly presents research topic and literature findings
10:20 – 11:05 AM	Session 1: <u>Research Methodology Design in ICT</u>	Theory /Lecture + Examples	Research design, experimental methods, simulations, data collection, validation techniques
11:05 AM – 12:15 PM	Session 2: Group Methodology Design Workshop	Hands-on /Collaborative Design	Groups prepare objectives, hypotheses, conceptual framework, tools/platforms, methodology flowchart
12:15 – 1:15 PM	Lunch Break		
1:15 – 2:30 PM	Session 3: <u>Scientific Paper Writing</u>	Theory /Lecture + Demonstration	Structure of scientific papers, abstract and article writing, publication ethics, plagiarism, journal/conference selection
2:30 – 2:45 PM	Tea/Coffee Break		
2:45 – 4:15 PM	Session 4: Group Proposal Writing Lab	Hands-on /Guided Writing	Groups prepare abstract, introduction, methodology section, proposal objectives, expected outcomes
4:15 – 5:00 PM	Proposal Clinic and Trainer Feedback	Interactive /Group Mentoring	Trainer reviews methodology and paper drafts with each group

Deliverables-

Each group prepares research methodology framework, draft proposal structure, draft scientific paper sections- Title, Abstract, Introduction, Methodology, and preliminary presentation slides

Day 3: Research Communication and Final Presentation

Theme: *From Conducting Research to Communication*

Time	Session	Type/Method	Topics / Activities
10:00 – 10:15 AM	Day 2 Recap & Presentation Planning	Interactive/Discussion	Final guidance and presentation preparation
10:15 – 11:00 AM	Session 1: <u>Conducting ICT Research and Publishing Outcomes</u>	Theory/Lecture + Case Studies	Research execution, data interpretation and publication workflow
11:00 AM – 12:15 PM	Session 2: Group Proposal and Paper Finalization Workshop	Hands-on /Group Work + Trainer Rotation	Final editing of proposal, refining paper sections, preparing final presentation
12:15 – 1:15 PM	Lunch Break		
1:15 – 2:30 PM	Session 3: <u>Research Communication and Technical Presentation Skills</u>	Theory /Demonstration + Practice	Effective technical presentation, proposal defense, responding to reviewers and evaluators
2:30 – 2:45 PM	Tea/Coffee Break		
2:45 – 4:30 PM	Session 4: Group Research Proposal & Paper Presentations	Hands-on/Group Presentation	Each group presents research proposal draft followed by feedback
4:30 – 5:00 PM	Closing Session & Certificate Distribution	Closing /Reflection & Networking	Workshop summary, feedback, future collaboration and publication opportunities